

Lecture - 8 - Data splits, models & cross-validation

- Bias / Variance
- Regularization
- Train / dev / test splits
- Model selection & Cross Validation.

If data underfits, it is said to have high bias.

If data overfits, it is said to have high variance.

Regularization is used to prevent overfitting of data.

For linear regression

$$\sum_{i=1}^n \|y^{(i)} - \theta^T x^{(i)}\|^2 + \frac{\lambda}{2} \|\theta\|^2$$

↓
regularization

For logistic regression.

$$\arg \max_{\theta} \sum_{i=1}^n \log(p(y^{(i)} | x^{(i)}; \theta)) - \frac{\lambda}{2} \|\theta\|^2$$

↓ Training set
 $S = \{(x_i, y_i)\}_{i=1}^m$

$$P(\theta|S) = \frac{P(S|\theta) \cdot P(\theta)}{P(S)}$$

$$\arg \max_{\theta} P(\theta|S) = \arg \max_{\theta} P(S|\theta) P(\theta)$$

$$\arg \max_{\theta} \left(\prod_{i=1}^m P(y^{(i)} | x^{(i)}; \theta) \right) P(\theta)$$

↑
 Logistic Regression.

$$P(\theta): \theta \sim \mathcal{N}(0, \tau^2 I)$$

$$P(\theta) = \frac{1}{\sqrt{2\pi}} | \tau^2 I |^{1/2} \exp \left(-\frac{1}{2} \theta^T (\tau^2 I)^{-1} \theta \right)$$

↓
 After doing the math, we reach to the regularized term.

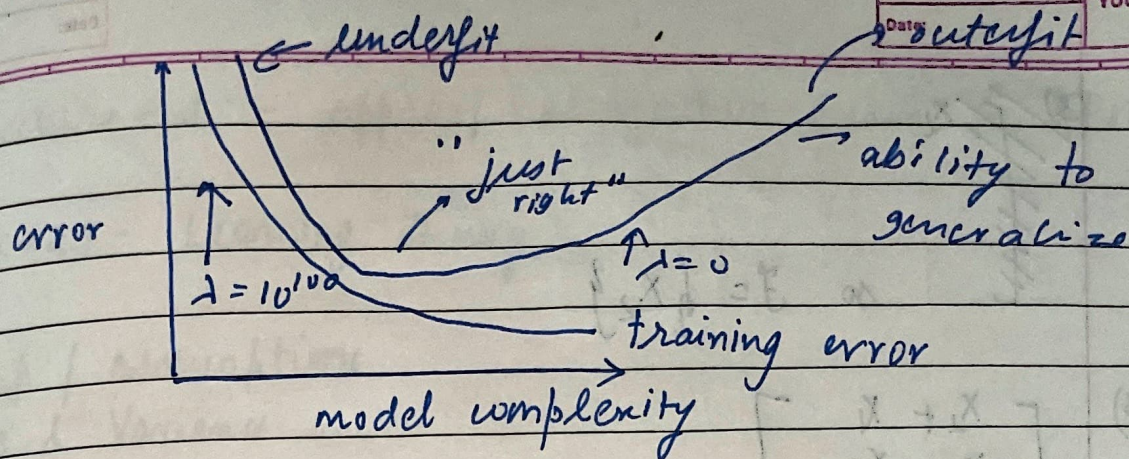
Frequentist

$$\arg \max_{\theta} P(S|\theta) - \text{MLE}$$

Bayesian

Prior distribution $P(\theta)$

$$\arg \max_{\theta} P(\theta|S) -$$



Curve holds true for regularization as well.

Feature Selection

this a special case of model selection.

If we have many no. of features ($n \gg m$), and we suspect only a small number of features are "relevant" to the learning task.

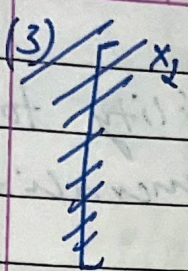
Start with $\mathcal{F} = \emptyset$

1) Then repeatedly try adding to $\mathcal{F}$ and see which single feature addition most improves dev set performance

2) Add feature to $\mathcal{F}$

1) $x_1 \dots x_5$
 $\mathcal{F} = [\emptyset] \rightarrow h_{\emptyset}(x) = \theta_0$

2) $\begin{bmatrix} \phi_1 + x_1 \\ \phi + x_2 \\ \vdots \\ \phi + x_5 \end{bmatrix} \rightarrow h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \dots + \theta_5 x_5$
 suppose x_2 improves performance the best.



so $y = f(X_2)$

(3)
$$\begin{bmatrix} X_2 + X_1 \\ X_2 + X_3 \\ \vdots \\ X_2 + X_4 \end{bmatrix}$$

$X_4 \checkmark$

$y = f(X_2, X_4)$